On Homological Mirror Symemtry

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The goal of our research is to prove homological mirror symmetry conjecture or to develop the relevant area of study. The claim of homological mirror symmetry conjecture is A_∞-equivalence of Fukaya category $Fuk(X)$ and A_∞-enhancement $D^b(Coh(X))$ (typically noted as $D^b(X)$) of derived category of coherent sheaves. This statement says the elaboration of mirror symmetry, originally correspondence of symplectic data on Hodge structure and that of complex in Calabi-Yau manifold, whose symplectic side we name as A-model, the complex side as B-model, mirror symmetry solves many of QFT problem as well as Landau-Ginzburg model and others, and homological mirror symmetry is generalization of mirror symmetry in ∞-category. The proof isn't easy, but the understanding of the problem statement itself is not difficult. I'll write the overview of the problem statement is schematically, precise but not very rigorous, and further I'll write the few possible speculation.

Definition 0.1. A_∞**-Tool Set**

(A_∞-Category)
An A_∞ category $\mathcal{A}$ is a quasi-category which consists of objects $X \in Ob(\mathcal{A})$ and 1-morphisms $f \in Mor^1(\mathcal{A})$, and the 1-morphism contains higher coherence $\alpha : f \Rightarrow g$, called 2-morphism, and iteratively n-morphisms.

(A_∞-Functor)
For A_∞ category $\mathcal{C}$ and $\mathcal{D}$, A_∞ functor $F : \mathcal{C} \to \mathcal{D}$ consists of $F_0 : Ob(\mathcal{C}) \to Ob(\mathcal{D})$ and higher maps $F_n : Hom_{\mathcal{C}}(X_{n-1}, X_n) \otimes \cdots \otimes Hom_{\mathcal{C}}(X_0, X_1) \to Hom_{\mathcal{D}}(F_0(X_0), F_0(X_n))$. F_0 is a mapping of objects, and $F_1(f)$ is a mapping of morphisms, but it doesn't satisfy the composition $(f \circ g) \circ h \ncong f \circ (g \circ h)$. Thus the higher map F_n where $n \geqq 2$ measures the failure of the composition.

Definition 0.2. Fukaya Category

For a symplectic manifold X, Fukaya category $Fuk(X)$ is an A_∞ category describing Lagrangian intersection. The objects of Fukaya category $Ob(Fuk(X))$ is Lagrangian submanifolds $L \subset X$, and the morphisms are Floer chain complex $CF^*(L_0, L_1)$ for each L_1 and L_2. This $CF^*(L_0, L_1)$ is a $\mathbb{Z}$-graded vector space defined by $CF^*(L_0, L_1) = \bigoplus_{p \in L_0 \cap L_1} \Bbbk p$. Floer chain complex is generated by its intersection points, and this is equivalent with counting critical points of Morse theory.

In Fukaya category, there is a composition $m_k : CF^*(L_{k-1}, L_k) \otimes \cdots \otimes CF^*(L_1, L_2) \to CF^*(L_0, L_k)[2-k]$ satisfying Stasheff associahedron property.

A_∞ enhancement of dg-category doesn't have unique enhancement, but the uniqueness is guaranteed if dg-category is derived category $D^b(X)$.

Definition 0.3. A_∞ **Derived Category**

A_∞ enhancement of derived categeory $D^b(X)$ is an A_∞ category $\mathcal{A}$, whose homology is $H^*(\mathcal{A}(E, F)) = Ext^*(E, F)$, while its trivialization $H^0(\mathcal{A}(E, F))$ needs to become the based derived category $H^0(\mathcal{A}(E, F)) \cong Hom_{D^b(X)}(E, F)$. This A_∞ enhancement $\mathcal{A}$ can be denoted as $D^b(X)$ by abuse of notation.

Accordingly, homological mirror symmetry conjecture claim that there is an A_∞ equivalence of categories where $D^b(Y)$ is an A_∞ enhanced derived category, and Y is mirror dual of X.

Conjecture 0.1. Homological Mirror Symmetry

The ordinary HMS conjecture is

$$Fuk(X) \cong D^b(Y)$$

Some modern version of HMS conjecture is A_∞ equivalence of dg-category of right hand side

$$Fuk(X) \cong Perf_{dg}(Y)$$

From the above statement of HMS conjecture, the left hand side $Fuk(X)$ describes the symplectic structure called A-model, and right hand side $D^b(X)$ describes complex structure called B-model. The spontaneous question is if there is any different variety of HMS conjecture, possibly more simplied version of that model, or some model already being verified? In particular, original version of mirror symmetry or the QFT version?

This might be a little derailed, but the history of mirror symmetry to catch up the intuition of what A/B-models are. The minimum version of mirror symmetry is the correspondence of Hodge structure of symplectic side called A-model and complex side called B-model. According to the outcome of Kähler geometry, symplectic data can be contained in Hodge diamond (why).

Definition 0.4. Mirror Symmetry – Classical Ver

For X the Calabi-Yau 3-fold,
$h^{1,1}(X) = h^{2,1}(Y)$
$h^{2,1}(X) = h^{1,2}(Y)$ TBD

The different version of mirror symmetry is QFT. From Marc Gross, classical intersection theory of Kontsevich moduli space whose quantization derives quantum cohomology, which makes semi-infinite variation of Hodge structure in A-model, while period integral of Landau-Ginzburg model generates SCFT, which makes another semi-infinite variation of Hodge structure in B-model. I only introduce A-model side:

Definition 0.5. Mirror Symmetry – QFT Ver

TBD – See Marc Gross "Tropical Geometry and Mirror Symmetry"

Going back to HMS, I'll continue making a paraphrase of the conjecture to make a strategy. From the conjectural formula $Fuk(X) \cong Perf_{dg}(Y)$, Fukaya category $Fuk(X)$ can be transformed to a stable ∞-category $tw^\pi(Fuk(X))$ called split-closed (idempotent-complete) derived Fukaya category, where tw is a twisted complex and π is Karoubi completion, while the right hand side $Perf(Y)$, a category of perfect complexes, which is also stable ∞-category. This is not 100% but quite an analogue of HMS. In addition, perfect complex is well-studied by the languauge of derived algebraic geometry, consider the reconstructivity of the geometry Y from the category $Perf(Y)$ (as an analogy of Tannakian duality), so the natural question is if there is any morphism $tw^\pi(Fuk(X)) \to Y$. As a result, HMS is modified to be:

Proposition 0.1. Tannakian Duality

There is a 1-to-1(why) correspondence of geometry Y and its category of perfect complexes $Perf(Y)$

$$Y \leftrightarrow Perf(Y)$$

Conjecture 0.2. HMS Modified

$$tw^{\pi}(Fuk(X)) \cong Perf(Y)$$

In addition, there exists a Sunami (my name) morphism

$$Sunami : tw^{\pi}(Fuk(X)) \to Y$$

Beginning Fukaya category from what is already known, consider cotangent bundle T^*X, which is a symplectic manifold, Nadler-Zaslow claims an analogue of HMS for the case of $Fuk(T^*X)$ using microlocal analysis.

Proposition 0.2. Nadler-Zaslow

There is an equivalence of category

$$Fuk(T^*X) \cong Sh_c(X)$$

where c means constructible not compactness.

Question 0.1

- Can we reduce $Sh_c(X)$ to $Coh(X)$ or $Perf(X)$ so we can treat it algebraic geometrically?
- Can the cotangent bundle T^*X be generalized to any other symplectic geometry? Does geometric quotient $T^*X//G$ act functorially on $Fuk(T^*X//G)$? What about fibration, toric variety, quiver variety, birational geometry, and any other geometric operations?

Conjecture 0.3

There exists a following commutative diagram

$$\begin{array}{ccc} Fuk(T^*X) & \longrightarrow & Sh_c(X) \\ & \searrow & \downarrow \\ & & Perf(X) \end{array}$$

Conjecture 0.4. Geometric Quotient in Fukaya Category

$$\begin{array}{ccc} T^*X & \longrightarrow & Fuk(T^*X) \\ \downarrow & & \downarrow \\ T^*X//G & \longrightarrow & Fuk(T^*X//G) \end{array}$$

Conjecture 0.5. Fiber Product Decomposition of Fukaya Category

Consider the category of constructible sheaves have decomposition by fiber product $Sh_c(X \times_Z Y) \cong Sh_c(X) \times_{Sh_c(Z)} Sh_c(Y)$, is there any Fukaya analogue $Fuk(X \times_Z Y) \cong Fuk(X) \times_{Fuk(Z)} Fuk(Y)$?

Hochschild cochain complex might be used for understanding Fukaya category, and just consider $HH^{\cdot}(Fuk(X))$ to induce GW invariants or Symplectic cohomology through open-closed mapping.

Definition 0.6. Hochschild Cochain Complex

The Hochschild cochain complex of a category $\mathcal{C}$ is $HH^{\cdot}(\mathcal{C}) \cong Ext^{\cdot}_{\mathcal{C}^e}(\mathcal{C},\mathcal{C})$ where $\mathcal{C}^e = \mathcal{C}^{op} \otimes \mathcal{C}$.

Considered that in particular shifted Hochschild complex with Gerstenhaber bracket is dgla, I define deformation functor for the study of deformation of dgla. The deformation functor of dgla spontaneously defines formal moduli problem, and I think it will be a wonderful argument, since Fukaya cagegory for example or any other categories has more structure than its Hochschild.

Definition 0.7. Quantum Cohomology

TBD

Definition 0.8. Symplectic Cohomology

TBD

Definition 0.9. Open-Closed Mapping

$$QH^*(X) \overset{CO}{\cong} HH_*(\mathcal{F}(X))$$

$$HH_*(\mathcal{F}(X)) \overset{OC}{\cong} QH^{*+n}(X)$$

where CO and OC are closed-open and open-closed map.

So far, Fukaya category has an GW invariant, but GW invariant doesn't describe the whole picture of Fukaya category. As stable ∞-category has a derived geometric interpretation according to Lurie, I'll consider Fukaya category $Fuk(X)$, and its transformation to stable ∞-category $Tw^{\pi}(Fuk(X))$.

Definition 0.10. Stable ∞-Category

A stable ∞-category $\mathcal{C}$ consits of zero objects $0 \in \mathcal{C}$ meaning it's object both initial and terminal. $\mathcal{C}$ has all finite limits and finite colimits. A commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

is a pullback iff it's a pushout.

Though it's different from $Tw^{\pi}(Fuk(X))$, some example of stable ∞-category is $\mathcal{C} = Perf(X)$, and the loop space $LX = Map(S^1, X)$ can also be derived self-intersection $LX = X \times^h_{X \times X} X$, connecting to Hochschild complex.

Definition 0.11. Connection to DAG

TBD

Meanwhile, I'll consider the formal scheme interpretation without using Hochschild cochain complex.

Definition 0.12. Formal Scheme

A formal scheme is a locally ringed space $(X, \mathcal{O}_X)$ that is locally isomorphic to formal spectrum $Spf(A)$ of a complete topological ring A. Assuming A is I-adically complete for an ideal $I \subset A$ s.t. $A = \varprojlim_n A/I^n$. Then as a set, $|Spf(A)| = \{\mathfrak{p} \subset A |\ \mathfrak{p} \text{ is open and contains } I^n \text{ for some } n\}$. As a remark, prime ideal $\mathfrak{p}$ is defined to be closed in Zariski topology, but from different point of view, it could be open subset in I-adic topology, different from Zariski.

For a scheme X, formal subscheme $\hat{X_L}$ of $L \subset X$ is given by a limit of sheaf $\hat{X_L} = \varprojlim_n \mathcal{O}_X/\mathcal{I}_L^n$ where $\mathcal{I}_L$ is an ideal sheaf of $\mathcal{O}_X$ for L.

For investigating few of the properties, like an ordinary scheme, a formal scheme has closed subschemes in terms of Zariski topology, and any proper close subscheme has codimension at least 1, and in addition the intersection of close subscheme has only finite points (why).

In particular, for the case of Fukaya category $Fuk(X)$, $CF^*(L,L)$ is A_∞ algebra, generating its formal scheme $Spf(Sym(CF^*(\hat{L}, L)^\vee[-1]))$. The differential structure d_Q was a direct consequence of coderivation $Q : T^c(CF^*(L,L)^\vee[-1]) \to T^c(CF^*(L,L)^\vee[-1])$, with A_∞ coherence.

Example 0.1. Non-Commutative Formal Pointed DG-Manifold

For coalgebra derivation $Q : T^c(V[1]) \to T^c(V[1])$ s.t. $Q^2 = 0$ of A_∞ structure where V is a vector space, there is a correspondence to formal scheme $(V, Q) \leftrightarrow (Spf(A), d_Q)$, where $V^\vee$ is a dual vector space, and $A = \hat{Sym}(V^\vee)$, and $d_Q : Spf(A) \to Spf(A)$ is a differential s.t. $d_Q(f) = Q(f)$ for every function $f \in A$.

So far ncfp dg-manifold has interpretation of formal geometry, and it was a domain of algebraic geometry. However, ncfp dg-manifold contains indispensable ingredient d_Q other than $Spf(A)$, not atually defined in formal scheme yet, but derived formal scheme includes differential d_Q data in itself. For defining derived formal scheme:

Definition 0.13. Derived Formal Scheme

Derived formal scheme is a functor $cdga \to \infty - groupoid$ s.t. $\pi_0(A) = Spf(A)$.

As an example, for the case of Fukaya category $Fuk(X)$, $CF^*(L,L)$ is A_∞ algebra, and it generates $Spf(Sym(CF^*(\hat{L}, L)^\vee[-1]$ Adding the derived structure, it will be (TBD).

So far we created derived formal scheme from Fukaya category in two ways: one uses Hochschild and the other is not. What is the relationship between them if any? So to summarize the above arguments, studying Fukaya category involves:

- Quantum/Symplectic cohomology
- Stable ∞-category
- Derived formal scheme

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1 A_∞-Category

What is the structure of A_∞ category? The natural question is geometric realization of A_∞ category, which might be Hochschild cochain complex and derive formal scheme, and I use the method of coalgebra construction for them. The geometric interpretation of A_∞-algebra is noncommutative formal pointed dg-manifold (hereafter in short, ncfp dg-manifold), and A_∞-category is generalization of A_∞-algebra. Indeed, similarly A_∞-category has a correspondence coalgebra, where we can define its geometry.

Definition 1.1. Counital Coalgebra

Coalgebra B is a graded vector space with morphism $\Delta : B \to B \otimes B$ s.t. $(\Delta \otimes id) \circ \Delta = (id \otimes \Delta) \circ \Delta$ called coproduct, and counital coalgebra is coalgebra B with unit $\epsilon : B \to 1$ where 1 is the unit object of $\mathcal{C}$, satisfying $(\epsilon \otimes id) \circ \Delta = (id \otimes \epsilon) \circ \Delta$. Counital coalgebra B is cofree counital coalgebra if it's given by $B = C(V) \cong T^c(V)$.

Definition 1.2. Non-Commutative Formal Pointed DG-Manifold

(A Non-Commutative Thin Scheme)
A non-commutative thin scheme X is a functor $X : Alg_{\mathcal{C}^f} \to Sets$, that is isomorphic to $X \cong Hom(-^*, B_X)$ for some coalgebra B_X, where $\mathcal{C}^f$ is artinian category of finite dimensional objects in $\mathcal{C}$, where $\mathcal{C}$ is $\mathbb{Z}$-graded vector spaces. In particular, X is a non-commutative formal manifold, if the functor X is isomorphic to some $Spc(T(V))$ for a cofree counital tensor coalgebra $T(V) = \underset{\geqq 1}{\oplus} V^{\otimes n}$. The dimension of X is defined as $dim_k V$.

(A Non-Commutative Thin DG-Scheme)
A non-commutative thin dg-scheme is a pair (X, d) where X is a non-commutative thin scheme, and d is a vector field on X of degree $+1$ satisfying bracket of dgla as $[d, d] = 0$. This d is called homological vector field.

(Non-Commutative Formal Pointed DG-Manifold)
A non-commutative formal pointed dg-manifold is a pair $((X, x_0), d_X)$ such that (X, x_0) is a non-commutative formal pointed graded manifold, and d_X is a homological vector field on X such that $d_X|_{x_0} = 0$.

Proposition 1.1. Taylor Decomposition of Graded Vector Field

For the case of formal manifold X, let the corresponding cofree counital coalgebra be $B_X \cong T^c(V)$, and the homological vector field be coderivation $d_X : B_X \to B_X$, that can have restriction by cofreeness, thus by using projection, $\pi \circ d_X : B_X \to V[1]$. If in addition, d_X can be decomposed degreewise for its linearity, as $d_X = \underset{n \geqq 1}{\Sigma} d_X^{(n)}$ for linear maps $d_X^{(n)} : V^{\otimes n} \to V[1]$.

Definition 1.3. Non-Unital A_∞ Algebra

A non-unital A_∞-algebra $A = (A, m_n)$ is a graded vector space A and coherence condition $m_n : A^{\otimes n} \to A[2-n]$ with Stasheff relation, whose geometric interpretation is the structure of a nc fp dg-manifold $((X, x_0), d_X)$ together with a coalgebra isomorphism $B_X \cong T(T_{x_0}X)$ with $x_0 : k \to B_X$.

Definition 1.4. Structure of A_∞ Algebra

The structure of A_∞ Algebra on $V \in Ob(Vect_k^{\mathbb{Z}})$ is given by coderivation d of degree $+1$ of the non-counital cofree coalgebra $T_+(V[1]) = \underset{n\geqq 1}{\oplus} V^{\otimes n}$ s.t. $[d, d] = 0$.

Proposition 1.2. Stasheff Relation

The condition of nc fp dg-manifold $((X, x_0), d_X)$ derives Stasheff relation of the A_∞ algebra, that is $m_n := d_X^{(n)}$ for $m_n : A^{\otimes n} \to A$ s.t. $\underset{r+s+t=n}{\sum}(-1)^{r+st} m_{r+1+t}(id^{\otimes r} \otimes m_s \otimes id^{\otimes t}) = 0$, whose degree is $|m_n| = 2 - n$. The Taylor decomposition of homological vector field $\pi \circ d_X = \underset{n\geqq 1}{\sum} d_X^{(n)}$ is degreewise by linearity, as $d_X^{(n)} : B_X^{(n)} \to B_X^{(1)}$.

Definition 1.5. Non-Commutative Formal n-Pointed DG-Manifold

Nc formal n-pointed dg-manifold $(\mathcal{X}, d_\mathcal{X}, \{pt_i\})$ is nc formal graded dg-manifold $(\mathcal{X}, d_\mathcal{X})$, and the points are $pt_i \in \mathcal{X}(k)$ for $1 \leqq i \leqq n$, and coderivation satisfies $d_\mathcal{X}|_{pt_i} = 0$, $d_\mathcal{X}|_{TX_i} : TX_i \to \underset{j,k}{\oplus} TX_j \oplus TX_k$ where X_i is formal neighborhood of X at pt_i.

Definition 1.6. A_∞-Category

A_∞-category $\mathcal{A}$ is nc formal dg-manifold $(\mathcal{X}, d_\mathcal{X}, \{pt_i\})$ with its coalgebra $B_\mathcal{X} \cong T^c(V[1])$. This V has a decomposition $V = \underset{i,j}{\oplus} E_{ij}$, where $E_{ij} := T(\widehat{X_i \times X}_{j\, pt_i, pt_j})$.

Remark 1.1. T

is V is a k-linear associative algebra generated from k-linear category $\mathcal{C}$: let $Ob(\mathcal{C}) = I$ to define an algebra $V = \oplus Hom(i, j)$ for $i, j \in I$ is a k-linear algebra with the unit $1_V = \underset{i\in I}{\sum} id_i$, where $\pi_i = id_i \in V$ is a commuting idempotent i.e. $\pi_i^2 = \pi_i$, $\pi_i\pi_j = \pi_j\pi_i$. This makes homomorphism $k^I \to V$ of algebra with unit.

Proposition 1.3. Stasheff Relation

From non-unital A_∞ category $\mathcal{A}$, the coalgebra $d_\mathcal{X} : T^c(\oplus E_{ij}[1]) \to T^c(\oplus E_{ij}[1])$ satisfies the Stasheff relation. Consider the tensor coalgebra can be given by the grading structure $T^c(\underset{1\leqq i,j\leqq n}{\oplus} E_{ij}[1]) = \underset{m\geqq 0}{\oplus} \underset{1\leqq i_0, i_1, i_2, \cdots i_m \leqq n}{\oplus} E_{i_0 i_1} \otimes E_{i_1 i_2} \otimes \cdots \otimes E_{i_{n-1} i_n} := \underset{m\geqq 0}{\oplus} B_\mathcal{X}^m$. From cofreeness of the coalgebra, the coderivation $d_\mathcal{X}$ canonically projected onto $\pi : B_\mathcal{X} \to B_\mathcal{X}^1$, and that can be decomposed linearwise, so $\pi \circ d_\mathcal{X} = \Sigma d_\mathcal{X}^n$ where $d_\mathcal{X}^n : B_\mathcal{X}^n \to B_\mathcal{X}^1$, and we have Stasheff relation from $[d_\mathcal{X}, d_\mathcal{X}] = 0$, and $m_n := d_\mathcal{X}^n$.

Remark 1.2

$d_\mathcal{X} \circ \eta = 0$ implies $d_\mathcal{X} \circ \eta(1) = 0$ or $d_\mathcal{X}(1) = 0$. This means curvature is $m_0 = 0$.

Remark 1.3

For the case of $I = \{i\}$ consists of one vertex, we write $d|_{pt} = d \circ \eta$.

For A_∞ category and its relationship with algebraic geometry, Kontsevich-Soibelman claims definition of ncfp dg-manifold of A_∞ algebra and its identification with Hochschild cochain complex as well as formal scheme. This notion can be generalized to A_∞ category including Fukaya category via coalgebra, and I don't know the prospective of future development, but connection of Stable ∞-category from Lurie might have connection to derived scheme.

On the other hand, there's a transformation of ncfp dg-manifold of A_∞-algebra into derived formal scheme, and the derived method captures the missing data d_Q used for A_∞ coherence, where ordinary formal scheme cannot have. In dag, we use cdga (A, d), consisting differential structure on a ring A.

Definition 1.7. Hochschild Cochain Complex

For an associative algebra A, take $A^e = A \otimes A$. Let M as A-bimodule or equivalenrtly A^e-module, we define Hochschild cochain as $CC^n(A, M) = Hom_{A^e}(A^{\otimes n}, M)$, and it's differential is given by $d_{Hoch} : CC^n \to CC^{n+1}$. Its cohomology class is given by homological extension $HC^\cdot(A, M) = Ext^\cdot_{A^e}(A, M)$.

For the case of category $\mathcal{C}$, the Hochschild cochain complex is $HH^\cdot(\mathcal{C}) \cong Ext^\cdot_{\mathcal{C}^e}(\mathcal{C}, \mathcal{C})$ where $\mathcal{C}^e = \mathcal{C}^{op} \otimes \mathcal{C}$.

It should be verified that notion could be well-defined on Fukaya category. Fukaya category has the corresponding coalgebra by Kontsevich-Soibelman, It could be identified with a coderivation complex.

Proposition 1.4. Cobar Construction

$$CC^\cdot(A, A) = CC^\cdot(A) \cong Coder(BA)$$

In particular, $CC^\cdot(Fuk(X)) \cong Coder(BFuk(X))$ where coalgebra of Fukaya category $Fuk(X)$ is $BFuk(X)$.

Considered that in particular shifted Hochschild complex with Gerstenhaber bracket is dgla, I define deformation functor for the study of deformation of dgla. The deformation functor of dgla spontaneously defines formal moduli problem, and I think it will be a wonderful argument, since Fukaya cagegory for example or any other categories has more structure than its Hochschild.

Definition 1.8. Deformation Functor

Let $mathfrakg$ be a dgla, and Art is a category of artinian rings. Deformation functor is a functor $Def_{\mathfrak{g}} : Art \to Set$ s.t.

$$Def_{\mathfrak{g}}(R) := \frac{\{\mathfrak{g} \otimes \mathfrak{m}_R|\}}{\text{gauge}}$$

satisfying Schlessinger axiom (TBD).

---

2 Algebraic Geometry

2.1 Deformation Theory

Deformation theory studies infinitesimal change of geometry to understand its structure. Perhaps, the most primitive deformation example is deformation retract:

Definition 2.1. Deformation Retract

For a topological space X and a subspace $A \subset X$, a continuous map $r : X \to A$ is a retraction if it's restriction to A $r|_A$ is the identity map on A.

Consider, in particular, the projection of tangent bundle to its base manifold $TM \to M$ is a deformation retract, where M is a smooth manifold, and its converse is generated from differentiating the possible path on each point, which makes tangent space. Let's make a tangent bundle:

Example 2.1. Tangent Bundle

A smooth manifold M, whose deformation is tangent bundle TM. From functor of point perspective, the tangent bundle should be $TM = Hom(spec(k[\epsilon]/(\epsilon^2), M)$. The mapping of Hom is a functional space s.t. $a + b\epsilon \mapsto f(a + b\epsilon)$ for each $f \in Hom(spec(k[\epsilon]/(\epsilon^2), M)$, and satisfying $f(a + b\epsilon) = f(a) + bf'(a)\epsilon$ by linear appximation by infinitesimality of ϵ. Verify $f(a) + bf'(a) \in TM$ where we put $g = f(a) \in M$ and $v = f'(a) \in T_gM$.

In the above example, I used an artinian algebra $k[x]/(x^2)$ to make a tangent bundle. By continuing the argument, can we apply the deformation technique to singular manifolds? For the case of algebraic variety, the tangent space is definable at every point regardless of singularity.

Example 2.2. Deformation – Singular Case

Let M be an algebraic variety. For $p \in M$, if the singular point is a closed point. the Zariski tangent space is defined by naïvely

$$T_pM = \mathfrak{m}/\mathfrak{m}^2 \text{where } \{p\} = V(\mathfrak{m})$$

Nonetheless, not all points are guaranteed to be singular. Consider

$$T_pM = Hom_{\kappa(p)}(\mathfrak{m}_p/\mathfrak{m}_p^2, \kappa(p)) \text{where } \kappa(p) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \text{ and } \mathfrak{m}_p = \mathfrak{p}A_{\mathfrak{p}}$$

Or more concisely,

$$T_xM \cong Hom_{k-alg}(\mathcal{O}_{M,x}, k[x]/(x^2))$$

Accordingly, the Zariski tangent space exists everywhere on M.

As a result of existence of Zariski tangent space at every point $p \in M$, the tangent sheaf $\mathcal{T}_M = Hom(\otimes_M, \mathcal{O}_M)$ cane be constructed from this data. Now if our goal of deformation is to create tangent bundle TM, and existence of TM depends only if the tangent sheaf is locally free, which isn't generally true. I'll consider cotangent complex to resolve this issue.

Definition 2.2. Cotangent Complex

A cotangent complex $\mathbb{L}_{B/A} \in D^b(B)$ is a chain complex s.t. $\mathbb{L}_{B/A} := \Omega^1_{P_\cdot/A} \otimes_{P_\cdot} B$ where $P_\cdot \to B$ is a simplicial resolution, and its dual $\mathbb{L}_{X/Y}$ where $X = Spec(A)$ and $Y = Spec(B)$. Often in short, $\mathbb{L}_X := \mathbb{L}_{X/k}$. The dual notion of cotangent complex is tangent complex $\mathbb{T}_{X/Y} = \mathcal{R}\mathrm{Hom}_{\mathcal{O}_X}(L_{X/Y}, \mathcal{O}_X)$.

Example 2.3. Deformation – Singular Case (Revisited)

Another approach of deformation theory is to make more than tangent bundle. If we use higher order $k[x]/(x^n)$ for $n \geqq 3$ or any artinian rings in general, it will generate the different geoemtric objects.

Deformation is a study of deformation functor,

Example 2.4. Deformation – Obstruction Bundle

Use Ext as obstruction instead of Zariski tangent space. TBD.

Finally, we will formalize the argument above into deformation functor.

Definition 2.3. Deformation Functor

$$Def_{\mathfrak{g}}(R) = \{x \in \mathfrak{g} \otimes \mathfrak{m}_R | dx + \frac{1}{2}[x, x] = 0\}$$

The problem of classical algebraic geometry is intersection theory problem. In affine scheme, the fiber product is a dual of tensor product.

Example 2.5

$$\begin{array}{ccc} X & \longrightarrow & ? \\ \downarrow & & \downarrow \\ k & \longrightarrow & k[x]/(x^2) \end{array}$$

Deformation guarantees the existence of tangent bundles if obstruction vanishes. What is the meaning of obstruction?

derived technique is a branch of algebraic geometry that can contain obstruction class.

observation 1: literal mean of deformation in algebraic topology (deformation retract)

observation 2: algebraic deformation

observation 3: geometric deformation

observation 4: categeorical deformation

observation 5: derived technique

One of the huge application of deformation theory is to solve moduli problem. Moduli space is roughly speaking a parameter space where each point corresponds to some object, whose deformation is an activity of generating neighborhood of moduli space, and this is often called moving of the point.

Assuming point x corresponds to $[X_0]$ in the moduli parameter space, define the movement of the point x is the deformation of the corresponding geometry $[X_0]$. Its deformation is the family $[X_t]$, which corresponds to the path $f(t)$ where $f(0) = p$, and $t \in U$ is the open neighborhood.

Question 2.1

Consider the deformation theory of moduli space, what is the invariant? For example, does deformation of a points x really belong to the moduli space $\mathcal{M}$ itself, so $U \subset \mathcal{M}$?

Perhaps, the solution is providing new spaces such as derived moduli space $\mathcal{M}^{der}$ or moduli of moduli space or any others.

Definition 2.4. Moduli Problem and Deformation

The deformation of point of derived moduli space $[E] \in \mathcal{M}^{der}$ where $E \to X$ is a vector bundle is controlled by $Ext^{\cdot}(E,E)$, that can be $T_{[E]}\mathcal{M}^{der} \cong Ext^{\cdot}(E,E)[1]$.

I'd also consider derived version of moduli sapce $\mathcal{M}^{der}$ for the sake of deformation invariant.

Definition 2.5. Derived Moduli Space

Derived moduli space $\mathcal{M}^{der}$ is a derived enhancement of moduli space $\mathcal{M}$. Derived enhancement of moduli space $\mathcal{M}$ at each point $p \in \mathcal{M}$ maps to tangent complex $p \mapsto T_p\mathcal{M}^{der}$, which makes a tangent sheaf structure, which needs to be locally free.

Another variety is formal moduli space, that can be generated from formal deformation of some space. Formal moduli space is that it's closed under formal deformation.

Definition 2.6. Formal Moduli Space

Formal moduli space is infinitesimal neighborhood of a point $x \in \mathcal{M}$ inside a moduli space $\mathcal{M}$. Formal moduli space is controlled by deformation of dgla.

$$\hat{\mathcal{M}}_E \leftrightarrow Def(\mathfrak{g})$$

Definition 2.7. Derived Formal Moduli Space

TBD

What about deformation problem in Fukaya category? The morphism of Fukaya category $CF(L,L)$ controls Lagrangian intersection data, which makes formal neighborhood $U = \hat{\mathcal{M}}_{[L]} = MC(CF^*(L,L))/\text{gauge}$.

Question 2.2. Fukaya Category and Formal Moduli

Consider relationship $\hat{\mathcal{M}}_{[L]} \to \hat{L}$ formal schemes. To find something meaningful TBD

2.2 Derived Algebraic Geometry

To list the problem of classical algebraic geometry.

- Multiplicity of zero points $y = x^2$
- Intersection if not transversal
- Moduli problem

Beginning from introducing few direct application of derived intersection.

Derived intersection resolves multiplicity and non-transversality in classical intersection. Derived intersection $X \times_Z^R Y$ is a derived version of fiber product $X \times_Z Y$, which is set theoretically an intersection $X \cap Y$. Consider that this notion can be reformulated in commutative diagram, and in category of topological spaces Top, all the morphisms are inclusion, and the pull back is $\iota_0, \iota_1 : X \cap Y \hookrightarrow X, Y$. Similarly, in category of affine schemes, pull back is $\iota_0, \iota_1 : X \times_Z Y \hookrightarrow X, Y$. The dual is commutative ring $X = Spec(A)$, $Y = Spec(B)$ and $Z = Spec(R)$ and A and B are R algebra, and $X \times_Z Y = Spec(A \otimes_R B)$, since $A \otimes_R B$ is still an R-algebra, and the commutative diagram below is the visualization:

$$\begin{array}{ccc} X \times_Z Y & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & Z \end{array} \qquad \begin{array}{ccc} A \otimes_R B & \longleftarrow & B \\ \uparrow & & \uparrow \\ A & \longleftarrow & R \end{array}$$

What's the missing data in this formalization is the multiplicatively and self-intersections. Consider the affine plane for the simplicity of argument and $A = B = k[x]$ and $R = k[x, y]$, the tensor product is $k[x] \otimes_{k[x,y]} k[x] = k[x]$, which means $\mathbb{A}^1 \otimes_{\mathbb{A}^2} \mathbb{A}^1 = \mathbb{A}^1$, so it's self-intersection $A \cap A = A$, which is not transversal, and I use derived intersection to measure the failing of transversality, by capturing tangent spaces and i-th higher data $H^i(X)$, whose intersection $H^i(Z) = H^i(X) \otimes_{H^i(S)} H^i(Y)$ defines the higher structure of Z, or in other words, consider the intersection of the two deformed spaces, think of the meaning if we are already aware of what deformation is(what). Accordingly, derived intersection is classical intersection together with infinitesimal data.

Definition 2.8. Derived Intersection

Derived intersection $X \times_Z^R Y$ is a derived tensor product $A \otimes_R^L B$ s.t. $X \times_Z^R Y = Spec(A \otimes_R^L B)$. Derievd tensor product $A \otimes_R^L B$ is a chain complex $P \cdot \otimes_R B$ with a projective resolution $P. \to A$.

Let's calculate the derived intersection. From the previous example $A = B = k[x]$ and $R = k[x, y]$, TBA.

Question 2.3

Is derived intersection invariant under deformation X_t and Y_t? It's generally no.

Remark 2.1. Local Complete Intersection

TBD

Example 2.6. Virtual Fundamental Class

The classical fundamental class is an algebraic cycle of homology class $[M] \in H_{2n}(M)$ for M a closed n-dimensional manifold, which is the identity by the intersection pairing $[M] \cdot [Z] = [Z]$ for its any subscheme $Z \subset M$. The intersection pairing is literally correspondence of multiplication of algebraic cycles to that of set-theoretical intersection $[X] \cdot [Y] = [X \cap Y]$

Similarly, virtual fundamental class is an algebraic cycle $[M]^{vir} \in H_{2vdim(M)}(M)$, where virtual dimension $vdim(M)$ is defined by $vdim(M) := (1-g)(dim(X) - 3) + \int_\beta c_1(T_X) + n$, not necessarily the same $vdim(M) \neq dim(M)$, and he virtual fundamental class is an identity under the multiplication $[M]^{vir} \cdot [Z]^{vir} = [Z]^{vir}$. The multiplication of the algebraic cycle is defined by derived interseection $[X]^{vir} \cdot [Y]^{vir} = [X \times_Z^R Y]^{vir}$.

Accordingly, GW/DT invariants can be defined by using virtual fundamental class. I'll explain the meaning of GW invariants in other time.

Example 2.7

- $$\int_{[\overline{M}_{g,n}(X,\beta)]^{vir}} \prod_{i=1}^{n} \psi_i^{a_i} ev_i(\gamma_i) \qquad \text{(GW Invariants)}$$

- $$\int_{[M_\beta(X)]^{vir}} 1 \qquad \text{(DT Invariants)}$$

3 On ∞-Category

3.1 Geometry of Category

Localization of category is really the same as localization of commutative ring in that it has a multiplicative inverse from its subset, and the multiplicative inverse makes idals of commutative rings trivial, and the category theory is the same. I'll consider localization of Fukaya category.

3.2 Homotopy Theory

Let's consider categorification of homotopy theory. For a topological space X, a homotopy is classically an isomorphism class $[f] \in \pi^n(X)$ where $f : S^n \to X$, whose equivalence class is given by existence of homotopy equivalence $H : S^n \times [0,1] \to X$ where $f(x) = H(x,0)$ and $g(x) = H(x,1)$ with continuous transformation. In this case, there is the equivalence $f \sim g$, also noted by $[f] = [g]$. Accordingly, for a category of topological space Top over X, homotopy category whose morphism consists of homotopy equivalence $[X, Y]$. Henceforth,

Definition 3.1. Na ïve Homotopy Category

Homotopy category is a category $Ho(Top)$ where its morphism $Hom_{Ho(Top)}(X,Y) = [X,Y]$ with any objects X and Y. the morphism $[f], [g] \in Ho(Top)$ have a composition $[f] \circ [g] = [f \circ g]$.

To modernize this notion, the functions $f, g \in Map(S^n, X)$ are elements in the functional space $Map(S^n, X)$, which is a topological space, and the equivalence $f \sim g$ is path connectedness in $Map(S^n, X)$. As a result, number of homotopically distinguishable elements are number of connected components, thus $\pi_n(X) = \pi_0(Map(S^n, X))$. Let's replace $X = S^n$ and $Y = X$ symbolically. By using ∞-category, homotopy category will be defined as follows:

Definition 3.2. Homotopy Category

For $\mathcal{C}$ an ∞-category, homotopy category is a category $h\mathcal{C}$ where its morphism $Hom_{h\mathcal{C}}(X,Y) = \pi_0(Map_{\mathcal{C}}(X,Y))$ with any objects X and Y.

The spontaneous question is the inverse problem if we can generate ∞-category from this homotopy category. Using a model category $\mathcal{C}$, take W a set of all weak equivalences, then the localization $\mathcal{C}[W^{-1}]$ is ∞-category.

Proposition 3.1. From Model to ∞

TBD

Definition 3.3. Stable ∞-Category

Stable ∞-category $\mathcal{C}$ is an ∞-category consists of

- zero object $0 \in \mathcal{C}$
- every morphism $f : X \to Y$ has a fiber/cofiber through diagrams $fib(f) \to X \to Y$ and $X \to Y \to cofib(f)$.
- every pullback is pushout.

Note: there are a variety of different formulation of stable ∞-category.

Definition 3.4. Stable ∞-Category (Modified)

Stable ∞-category $\mathcal{C}$ is an ∞-category consists of

- zero object $0 \in \mathcal{C}$
- all finite limit/colimit
- suspension functor $\Sigma : \mathcal{C} \to \mathcal{C}$ is an equivalence. Suspension Σ is $\Sigma X : cofib(X \to 0)$, whose inverse Ω is $\Omega X : fib(0 \to X)$

Now to introduce some examples of stable ∞-categories, $Tw^{\pi}(Fuk(X))$ generated from Fukaya category is stable ∞-category, and category of prefect complexes $Perf(X)$ is also stable ∞-category.

Example 3.1. Split-Closed Derived Fukaya Category

In the process of making $Tw^{\pi}(Fuk(X))$ from Fukaya category $Fuk(X)$, first of all $Tw(Fuk(X))$ adds

- shifts $L[k]$
- finite direct sum $L_1 \oplus L_2$
- mapping cones $Cone(f)$ for $f : L_1 \to L_2$
- twisted complexes

and the split-closure (the denoted by π) adds idempotent completion $E \cong im(e) \oplus im(1-e)$, where e is an idempotent $e : E \to E$ s.t. $e^2 = e$.

What is the actual meaning of split? Split means the invertivility of injection, but the geometric analogue is the deformation retract.

Example 3.2. Perfect Complex

$Perf(X)$ is a stable symmetric monoidal ∞-category, whose objects are perfect complexes. $E \in D_{qc}(X)$ is perfect if for any open affine $U \subset X$, the projective resolution $P^{\cdot} \to E$ is of finite length. It's closed under derived tensor product $E \otimes^{L} F \in Perf(X)$ for $E, F \in Perf(X)$

Now what do the homotopical proerties do? In particular, fib/cofib is.TBD.

Let's consider the connection of Stable ∞-category to derived scheme.

Proposition 3.2. Connection of Stable ∞ to DAG

$Spec(HH^{\cdot}(\mathcal{C}))$

Definition 3.5. Derived Scheme

Definition 3.6. Triangualted Category

4 Balmer Spectrum

If A_∞ category is symmetric monoidal category, its homotopy category is tensor triangulated category. In particular, tensor stable ∞-category's homotopy category is tensor triangulated category.

Definition 4.1. Tensor Triangulated Category

Triangulated cateory with monoidal structure.

Balmer spectum is a Zariski spectrum induced from tensor triangulated sturucture to define the geometric realization of triangulated category. A notable non-obvious example of tensor triangulated category is stable homotopy category.

Definition 4.2. Balmer Spectrum

For a tensor triangulated category $\mathcal{C}$, Balmer spectrum $Spc(\mathcal{C})$ is defined by

Definition 4.3. Stable Homotopy Theory

Definition 4.4. Stable Homotopy Category

Generalizating Balmer spectrum to ∞-language?
$Spc(\mathcal{C}) := Spc(h\mathcal{C})$

5 Quantization and Deformation

I don't know where to put this section to the introduction, and it might be a branch of main question of this pdf. Away from category, I go back to the basic of abstract algebra to discuss examples of associativity, in particular quantization and deformation.

Recall non-commtuative scheme from its coalgebra has an identification of A_∞ algebra/category. Consider a functor $X : Alg \to Coalg$, mapping an associative algebra $O(X)$ to a coalgebra B_X. Associative algebra is more general than unital commutative associative algebra $CRing$, and it's different from classical commutative scheme. Some idea of understanding noncommutative scheme is

Recall a commutative ring defines an affine scheme by its ring spectrum $X = V(\mathfrak{p})$ with its structure sheaf $\mathcal{O}_X$, and that can also be described functorially as $F : CRing \to Set$. Hereafter I focus on functorial point of view.

- Deformation
- Quantization

I don't know the general classification of associative algebra, but one important fact is that commutative algebra's quantization is associative. I wonder if there is any connection to the classical algebraic geometry or derived algebraic geometry in general.

Remark 5.1. Associative Algebra Example

To name some examples of associative algebra, $U(\mathfrak{g})$, $k[G]$, $M_n(k)$, Hecke algebra, path algebra, quiver algebra etc, and quantization of associtive algebra is associtive algebra.

The spontaneous question is what is the invariant under the quantization operation? Notice especially the boring part is the matrix ring $M_n(k)$, since it's deformation invariant so it's Morita equivalent to k, but I'd also wonder the quantized version.

Definition 5.1. Morita Equivalence of Quantization

TBD

Question 5.1

Should we just use deformation of associative algebra, which makes A_∞ algebra?

Another approach is deformation (TBD).

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